

SEAT No. _____

No. of printed pages: 03

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SARDAR PATEL UNIVERSITY
Master of Science (Information Technology)
SEMESTER-I

PS01CINT34 - (Relational Database Management Systems)

25th February, 2021

Marks: 70

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Time: 10 AM To 12 Noon

Q1. Do as directed:

A. Select the Appropriate Option of the following questions as answer:

1. Column of the relation is referred as _____.
a. Record b. Attribute c. Tuple d. None of given
2. _____ constraint establishes uniqueness of records in a table.
a. Primary key b. CHECK c. Foreign key d. None of given
3. The _____ command is used to modify the structure of a table.
a. ALTER TABLE b. UPDATE c. MODIFY TABLE d. None of given
4. In Hierarchical Data Model, data are organized in the form of _____.
a. Tree b. Graph c. Relation d. None of given
5. _____ will be automatically fired on occurrence of predefined event.
a. Procedure b. Trigger c. Function d. None of given
6. _____ declares a variable to have the same data type as that of a previously defined variable or a column in the table.
a. Variable b. %type c. Declare d. None of given
7. When one instance of an entity is associated with more than one instance of an other entity then that relationship, is called
a. 1:1 b. 1:n c. m:n d. None of given
8. _____ cursor attribute returns number of records in a cursor.
a. %count b. %rowcount c. %Recordcount d. None of given

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B. Do as directed:

I Fill in the blanks:

- i. Full form of RDBMS is _____.
- ii. _____ command is used to retrieve the records from a table.
- iii. _____ command is used to take back a permission from the user.
- iv. The data can be shared by a single or multiple users. (True/False): _____.
- v. Full form of PL/SQL is _____.
- vi. In _____ Independence rule, the user is isolated from physical method of storing and retrieving information from the database.
- vii. A relation is called in _____ if there is no transition dependency exists.
- viii. The data stored in the cursor is called _____.

II Write answer in ONE word:

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- i. Which type of attributes are further divided into tree like structure?
- ii. Which symbol is used to represent entity in E-R diagram?
- iii. Where the procedures & functions are stored?
- iv. Which statement is used to transfer control to given block name?
- v. In which relationship, a relationship exists between occurrences of the same entity set?
- vi. Which built in function is used to find the length of a given string?
- vii. A query within a query is known as _____.
- viii. Which clause is used to impose condition on output given by GROUP BY clause?

Q2 Answer the following questions: [Any SEVEN]

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- i. List out any four types of database users
- ii. Write a command to display the result of $500 * 200$.
- iii. Define the term 'Data Sharing' and 'Data Protection'.
- iv. Differentiate clearly logical data independence and physical data independence.
- v. List out at least 4 the most common data types available in PL/SQL.
- vi. Write the disadvantages of SQL.
- vii. State the two differences between database Trigger & Stored procedure.
- viii. Define: Cardinality.
- ix. List out the responsibility of DBA.

Q3. What is DBMS? Give few examples of popular DBMS software. Write advantages and disadvantages of DBMS.

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OR

Q3. What is Database Model? List different types of database models and explain ANY ONE of them in detail.

Q4. Do as directed.

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- i. Explain briefly with example 1NF and 2NF.
- ii. Write short note on types of attributes

OR

Q4. Do as directed.

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- i. List out different symbols used to draw E-R diagram with its purpose.
- ii. Write short note on types of relationship

Q5. Write SQL command for each of the following. [Any FOUR]

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- i. To create a table EMP having fields (ENO number(3), DNO number(1), ENAME varchar2(15)), where ENO is primary key
- ii. List out DNO and DNO wise total BASIC_PAY for all employees of EMP table.
- iii. Give permission to modify the structure of table EMP to user XYZ.
- iv. Retrieve all the information of table EMP in ascending order of JOIN_DATE.
- v. Delete all the records of a table EMP.

OR

- Q5. Do as directed.** 4
i. Write short note on types of join. 2
ii. Define Sequence & View. 2
iii. Explain with example the built in function: ROUND() 8
- Q6. Do as directed.**
i. Explain briefly PL/SQL block structure with purposes of each section.
ii. Write short note on cursor including its types & attributes.

OR

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- Q6. Do as directed.**
i. Write a Procedure named PSUB, which subtracts to given numbers say M and N. Also write PL/SQL code block to read two numbers A and B and using above procedure PSUB subtract two numbers A and B and print the result.
ii. Explain briefly following statement with example of your choice: [Any ONE]
a. WHILE loop b. IF Statement
